

Obstruction Size versus Representation Blow-Up:

Rectangle Covers, Fooling Sets, and the Refutation of an Equality

A literature review and annotated bibliography, oriented toward the Lean mechanization in
`Ste.RepresentationBounds`

Project `thejeb`

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Abstract

The project’s open-question queue carried the item “obstruction size = representation blow-up”: the conjecture that the concrete Čech obstruction number of `Ste.CechObstruction` — the count of compatible-but-stuck families of local sections for the singleton cover — *equals* the representational cost that a coupled constraint forces on any variable-separable (rectangular) encoding. This note settles the item as a *negative* result with a precise positive replacement. There are two readings of “blow-up”. Under the *box-gap* reading the equality is true but tautological: the obstruction is by definition $|\text{box}(T)| - |T|$, the overshoot of the smallest single rectangle containing T . Under the *complexity* reading — the minimum number $\rho(T)$ of rectangles whose union is exactly T , the direct analogue of the nondeterministic cover number of communication complexity — the equality is *false in both directions*, and what is true is a sandwich of bounds: $\text{fool}(T) \leq \rho(T) \leq |T|$, with $\rho(T) = 1$ exactly when the Čech obstruction vanishes, and $\rho(\text{allEqual}_{n,k}) = k$ exactly while the obstruction is $k^n - k$. All of these statements, including both strict-separation witnesses refuting the equality, are machine-checked in Lean 4 in `Ste.RepresentationBounds`. This review surveys the communication complexity and extension-complexity literature that the definitions are drawn from, and states precisely which fragments are mechanized. Following the project rule — distrust results that are not machine proven — all claims about the mechanization are limited to statements that compile in CI; everything else is attributed to the cited literature.

1 The question, and why it is two questions

Let V be a set of variables with value types $(A_v)_{v \in V}$ and let $T \subseteq \prod_v A_v$ be a constraint (a hypothesis set in the set-theoretic-estimation reading). `Ste.CechObstruction` computes, for the singleton cover of V , the set of *compatible families* of local sections — which, because singleton contexts have empty overlaps, is exactly the box

$$\text{box}(T) = \prod_{v \in V} \pi_v(T),$$

the product of the per-variable projections — and defines the obstruction number

$$\text{obs}(T) = |\text{box}(T)| - |T| \in \mathbb{N} \cup \{\infty\},$$

the count of phantom assignments that margin-based reasoning must entertain and the coupling then kills. The open item asked whether $\text{obs}(T)$ *equals* the “forced representation blow-up” of T . The answer depends entirely on what a representation is.

Reading 1: blow-up of the single-rectangle representation. If a “representation” is one rectangle (one product set) and “blow-up” is how many spurious points the best such representation adds, then the best rectangle containing T is $\text{box}(T)$ itself, and the blow-up is $|\text{box}(T)| - |T| = \text{obs}(T)$ *by definition*. The Lean statement (`cechObstruction_eq_box_sub_encard`, with `compatibleFamilies_eq_boxProduct` exhibiting the box) is proved by unfolding two definitions. The equality holds, but it is bookkeeping, not a representation theorem; nothing in it deserves the name “blow-up equality”.

Reading 2: blow-up as the size of the smallest exact rectangular representation. The substantive reading, standard since Aho et al. [2] and systematized by Kushilevitz and Nisan [6], measures a constraint by its *rectangle-cover complexity*

$$\rho(T) = \min \left\{ m : T = R_1 \cup \dots \cup R_m, \ R_i \text{ rectangles} \right\},$$

the minimum number of product sets whose union is exactly T (each $R_i \subseteq T$ automatically). In two-party communication complexity $\log_2 \rho$ is the nondeterministic complexity of the predicate T up to one bit; in the extension-complexity literature [5, 7] ρ is the combinatorial lower bound (the rectangle covering bound) on nonnegative rank and hence on the size of linear-programming reformulations. Under this reading the conjectured equality $\text{obs}(T) = \rho(T)$ (or any uniform inequality between the two) is *false*, and the correct picture is the sandwich of Section 2.

2 The sandwich, as mechanized

All results in this section are machine-checked, sorry-free, in `lean/Ste/RepresentationBounds.lean`; names in typewriter font are the Lean identifiers. ρ is `rectCoverNumber`, valued in $\mathbb{N} \cup \{\infty\}$ (∞ if no finite cover exists); fool is `foolingNumber`, the supremum of cardinalities of fooling sets.

Definition 1 (fooling set; `IsFoolingSet`). $F \subseteq T$ is a fooling set for T if no rectangle $R \subseteq T$ contains two distinct points of F .

This is the abstract form of the classical definition: the usual condition — for $f \neq g$ in F some coordinate mixture of f and g leaves T — implies it, because rectangles are closed under coordinatewise mixing (`IsRectangle.mix_mem`), which is the mechanized engine of the argument.

Theorem 1 (lower bound; `IsFoolingSet.encard_le_of_hasRectCover`, `foolingNumber_le_rectCoverNumber`). If F is a fooling set for T , then $|F| \leq \rho(T)$; hence $\text{fool}(T) \leq \rho(T)$. Each rectangle of an exact cover is a sub-rectangle of T and therefore contains at most one fooling point; the mechanized proof builds the induced injection $F \rightarrow \{1, \dots, m\}$.

Theorem 2 (upper bound; `rectCoverNumber_le_encard`). $\rho(T) \leq |T|$: singletons are rectangles (`isRectangle_singleton`), so T covers itself pointwise. (For infinite T the right side is ∞ and the bound is vacuous.)

Theorem 3 (bottom of the scale; `rectCoverNumber_eq_one_iff`, `rectCoverNumber_eq_one_iff_cechVanishes`, `one_lt_rectCoverNumber_iff`). $\rho(T) = 1$ iff T is a nonempty rectangle; for nonempty T , $\rho(T) = 1$ iff the Čech obstruction vanishes, and $\rho(T) \geq 2$ iff it does not. (Via `cechVanishes_iff_rectangular` of `Ste.CouplingLowerBound`.) At the bottom of the scale, obstruction and cover complexity carry the same information — this, and only this, survives of the conjectured equality.

Theorem 4 (the coupling, exactly; `isFoolingSet_allEqual`, `rectCoverNumber_allEqual`, `foolingNumber_allEqual`). For the n -fold coupling `allEqualn,α` (all coordinates equal over an alphabet of size k) with $n \geq 2$: the k constant vectors form a fooling set, and the sandwich pinches: $\text{fool} = \rho = k$. Meanwhile $\text{obs}(\text{allEqual}_{n,\alpha}) = k^n - k$ (`cechObstruction_allEqual`, proved previously).

Theorem 5 (refutation of the equality, both directions; `rectCoverNumber_lt_cechObstruction_allEqual`, `exists_rectCoverNumber_lt_cechObstruction`, `exists_cechObstruction_lt_rectCoverNumber`, `not_forall_cechObstruction_le_rectCoverNumber`, `not_forall_rectCoverNumber_le_cechObstruction`). No uniform inequality holds between obs and ρ :

- every nonempty rectangle has $\text{obs} = 0 < 1 = \rho$ (witness: the full square on one Boolean variable), so $\rho \leq \text{obs}$ fails;
- for $n \geq 3$, $k \geq 2$, $\rho(\text{allEqual}_{n,\alpha}) = k < k^n - k = \text{obs}$ (witness: `allEqual3,Bool` with $2 < 6$), so $\text{obs} \leq \rho$ fails.

Remark 1 (the origin of the conjecture; `diagonal_eq_allEqual`, `rectCoverNumber_diagonal`, `diagonal_cechObstruction_eq_rectCoverNumber`). At the minimal nonvanishing instance — the two-variable Boolean diagonal, which is `allEqual2,Bool` — the two invariants coincide: $\text{obs} = 2^2 - 2 = 2 = \rho$. The equality conjecture is the smallest example mistaken for a law: it fails at $n = 3$ (6 vs. 2) and in the opposite direction for every nonempty rectangle (0 vs. 1).

The conceptual moral matches the communication-complexity literature: obs is a *volume* invariant (how much the box overshoots), ρ is a *covering* invariant (how many boxes are needed); volume bounds covering only through logarithms, never linearly. The classical relation — $\log_2 \rho$ equals nondeterministic communication complexity up to an additive bit, and $\text{fool}(T)$ can be exponentially smaller than $\rho(T)$ [4] — lives one abstraction level above what is mechanized here and remains outlook.

3 Annotated bibliography

Kushilevitz and Nisan [6] The standard monograph on two-party communication complexity; Chapters 1–2 define combinatorial rectangles, monochromatic covers, the cover numbers C^0, C^1 , and the fooling-set and rank lower-bound methods. Our $\rho(T)$ is precisely their C^1 read on the constraint T as a $|V|$ -dimensional 0/1 tensor (the book treats $|V| = 2$; the rectangle and fooling formalism, and our Lean proofs, are arity-uniform). The book’s Proposition 2.2 — every cover rectangle contains at most one fooling point — is exactly the statement mechanized as `IsFoolingSet.encard_le_of_hasRectCover`; the book’s log-scale accounting ($N^1(f) = \log_2 C^1(f)$) is the part we deliberately do not mechanize, since the STE obstruction is counted on the linear scale.

Aho et al. [2] The paper that introduced rectangle covers and the fooling-set bound in the VLSI/communication setting, and posed the question (later resolved negatively) of whether cover number and deterministic complexity are polynomially related. Historically this is where “representation size of a relation = number of monochromatic rectangles” enters the literature, which is why we take it as the anchor for the complexity reading of “blow-up”. The fooling-set method appears here in the concrete mixture form whose abstract closure property is our `IsRectangle.mix_mem`.

Yannakakis [7] Connects rectangle covers to representation size in the strongest classical sense: the size of the smallest linear program expressing a combinatorial polytope (its extension

complexity) is governed by the nonnegative rank of a slack matrix, of which the rectangle covering number is the support lower bound; symmetric LPs for matching and TSP are shown exponential. For the STE reading, this is the license to call ρ a “representation blow-up”: a constraint with $\rho(T) = m$ is exactly a constraint expressible as a disjunction of m variable-separable pieces and no fewer. Our $\rho(\text{allEqual}) = k$ theorem is a (very small) instance of the slack-support paradigm: the fooling set plays the role of a combinatorially independent family in the slack matrix.

Dietzfelbinger et al. [4] Quantifies the looseness of the fooling-set method against the rank method: fooling sets can be quadratically smaller than deterministic complexity bounds, and (via a counting argument) almost all functions have maximal complexity but only polynomial-size fooling sets. Relevant here as a caution: our sandwich $\text{fool}(T) \leq \rho(T) \leq |T|$ pinches to equality on `allEqual`, but the left inequality is in general far from tight, so `foolingNumber` should not be mistaken for a characterization of ρ . Any future STE conjecture of the form “fooling number = cover number for coupling-like constraints” must be tested against the separations in this paper.

Fiorini et al. [5] The modern culmination of Yannakakis’s program: unconditional exponential lower bounds on the extension complexity of the cut, stable set, and TSP polytopes, obtained by lower-bounding the rectangle covering/nonnegative-rank of explicit slack matrices (building on the correlation polytope and quantum communication arguments). For the STE project this fixes the right asymptotic vocabulary — obstructions to small rectangular representations are proved by exhibiting structured sub-matrices, not by counting phantom assignments — reinforcing the present verdict that `obs`, a phantom count, cannot equal a covering complexity in general.

Abramsky and Brandenburger [1] The sheaf-theoretic frame the STE Čech obstruction is modeled on: empirical models over a cover of contexts, with non-contextuality exactly the existence of a global section gluing the local ones. Our `obs(T)` is the crudest numerical shadow of their obstruction — the count of compatible-but-stuck families for the singleton cover — and the mechanized equivalence “`obs` vanishes iff T is a rectangle iff $\rho(T) = 1$ ” is the STE analogue of their “non-contextual iff product structure” at the coarsest cover. The paper’s cohomological refinements (with `Ste.CechCover`, `Ste.CechComplex`, `Ste.TwistedCech`) are where a *graded* obstruction-vs-cover-number comparison would have to live; nothing at that level is claimed here.

Dechter [3] The constraint-processing side of the same divide: bounded induced width buys a linear-size faithful junction-tree representation (mechanized earlier in `Ste.JunctionTree`), while `allEqual`, of maximal width, is exactly the constraint whose margins carry no information. The present note sharpens the hard side of that dichotomy: the coupling’s true rectangular representation cost is k (small!), and what explodes is not ρ but the box gap $k^n - k$ — the price of *margin-only* reasoning, not of *rectangular* reasoning. This distinction was invisible while the open item conflated the two.

4 Verdict and open questions

Verdict. The open item “obstruction size = representation blow-up” is *closed, negatively*, with the following machine-checked replacement. (i) As a box-gap statement the equality is a definition, not a theorem (`cechObstruction.eq.box.sub.encard`). (ii) As a statement about rectangle-cover

complexity it is false in both directions: $\text{obs} = 0 < \rho = 1$ on nonempty rectangles, and $\rho = k < k^n - k = \text{obs}$ on $\text{allEqual}_{n,k}$ for $n \geq 3, k \geq 2$ (`not_forall_*`). (iii) What is true and now mechanized is the sandwich $\text{fool}(T) \leq \rho(T) \leq |T|$, the boundary agreement $\rho(T) = 1 \Leftrightarrow \text{obs}(T)$ vanishes, and the exact value $\rho(\text{allEqual}_{n,\alpha}) = |\alpha|$ via the constant-vector fooling set. The equality holds at exactly one place we know of — the 2×2 diagonal, $\text{obs} = \rho = 2$ — and that coincidence is the likely origin of the conjecture.

Open questions for the queue. (1) *Logarithmic reconciliation*: is there a uniform bound $\rho(T) \leq f(\text{obs}(T), |T|)$, e.g. polynomial in $\log(\text{obs} + 2)$ and $\log(|T| + 2)$, for finite constraints? The allEqual family is consistent with $\rho \leq \text{obs} + 1$ for *nonempty non-rectangular* T — is $\rho(T) \leq \text{obs}(T) + 1$ always true there? (The fooling bound gives no obstruction to it; a proof or counterexample is a concrete next target.) (2) *Partition versus cover*: the disjoint-cover (partition) number $\rho^\sqcup$ satisfies $\rho \leq \rho^\sqcup$; does allEqual also have $\rho^\sqcup = k$ (the singleton-of-constants partition gives $\leq k$, and the fooling bound gives $\geq k$ — this should mechanize with the same machinery). (3) *Graded comparison*: for covers with nonempty overlaps (`Ste.CechCover`) is there a cover-relative ρ for which the vanishing boundary ($\rho = 1$) again matches the cover-level sheaf condition?

References

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